

2.2 The Inverse of a Matrix

I

Recall that the multiplicative inverse of a number such as 3 is $\frac{1}{3}$ or 3^{-1} . This inverse satisfies the equations

$$3^{-1}(3) = 1 \text{ and } 3(3^{-1}) = 1$$

The matrix generalization requires both equations and avoids the slanted-line notation for division because matrix multiplication is not commutative. Furthermore, a full generalization is possible only if the matrices involved are square. (See footnote in textbook p. 135)

An $n \times n$ matrix A is said to be invertible if there is an $n \times n$ matrix C such that

$$CA = I \text{ and } AC = I$$

where $I = I_n$, the $n \times n$ identity matrix. In this case, C is an inverse of A . In fact, C is uniquely determined by A , because if B were another inverse of A , then $B = BI = B(AC) = (BA)C = IC = C$.

This unique inverse is denoted A^{-1} , so that

$$\boxed{A^{-1}A = I \text{ and } AA^{-1} = I}$$

A matrix that is not invertible is sometimes called a singular matrix, and an invertible matrix is called a nonsingular matrix.

2.2 The Inverse of a Matrix (cont'd)

II

Here is a simple formula for the inverse of a 2×2 matrix, along with a test to tell if the inverse exists.

Theorem 4

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - cb \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - cb} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - cb = 0$, then A is not invertible.

Example 2

Find the inverse of $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$

Solution

Since $\det A = 3 \cdot 6 - 5 \cdot 4 = -2 \neq 0$, A is invertible.

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 6 & -4 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}$$

Theorem 5

If A is an $n \times n$ invertible matrix, then for each $\vec{b}$ in $\mathbb{R}^n$, the equation $A\vec{x} = \vec{b}$ has the unique solution $\vec{x} = A^{-1}\vec{b}$.

Proof

Take any $\vec{b}$ in $\mathbb{R}^n$. A solution exists because if $A^{-1}\vec{b}$ is substituted for $\vec{x}$, then

$$A\vec{x} = A(A^{-1}\vec{b}) = (AA^{-1})\vec{b} = I\vec{b} = \vec{b}$$

So $A^{-1}\vec{b}$ is a solution. To prove that the solution is unique, show that if $\vec{u}$ is any solution, then $\vec{u}$, in fact, must be $A^{-1}\vec{b}$. Indeed, if

$A\vec{u} = \vec{b}$, we can multiply both sides by A^{-1} and obtain

$$A^{-1}A\vec{u} = A^{-1}\vec{b}, \quad I\vec{u} = A^{-1}\vec{b}, \quad \text{and} \quad \vec{u} = A^{-1}\vec{b}$$

For me, this is more understandable:

$A\vec{x} = \vec{b}$; multiply both sides by A^{-1} from left; $A^{-1}A\vec{x} = A^{-1}\vec{b}$;
 $A^{-1}A = I$; $I\vec{x} = A^{-1}\vec{b}$; $I\vec{x}$ is the same as $\vec{x}$; $\vec{x} = A^{-1}\vec{b}$.

2.2 Inverse of a matrix (cont'd)

III

Example 3

A horizontal elastic beam is supported at each end and is subject to forces at points 1, 2 and 3, as shown in figure 1.

Let $\vec{f}$ in $\mathbb{R}^3$ list the forces at these points, and let $\vec{y}$ in $\mathbb{R}^3$ list the amount of deflection of the beam at these three points. Using Hooke's law from physics, it can be shown that

$$\vec{y} = D\vec{f}$$

where D is a flexibility matrix. Its inverse is called the stiffness matrix. Describe the physical significance of the columns of D and D^{-1} .

Solution

Write $I_3 = [e_1 \ e_2 \ e_3]$ and observe that

$$D = DI_3 = [De_1 \ De_2 \ De_3]$$

Interpret the vector $\vec{e}_1 = (1, 0, 0)$ as a unit force applied downward at point 1 on the beam (with zero force at the other two points). Then De_1 , the first column of D , lists the beam deflection due to a unit force at point 1. Similar descriptions apply to the second and third columns of D .

To study the stiffness matrix D^{-1} , observe that the equation $\vec{f} = D^{-1}\vec{y}$ computes a force vector $\vec{f}$ when a deflection vector $\vec{y}$ is given.

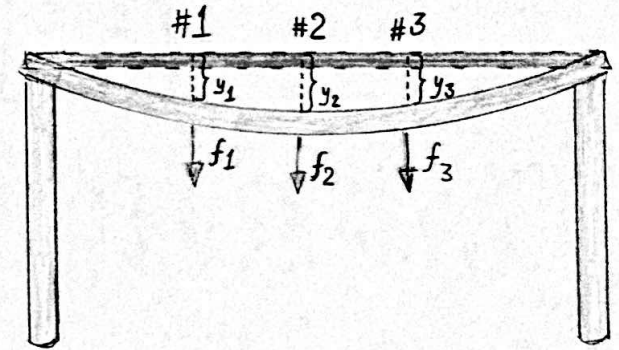


Figure 1 Deflection of an elastic beam.

Write

$$D^{-1} = D^{-1}I_3 = [D^{-1}\vec{e}_1 \ D^{-1}\vec{e}_2 \ D^{-1}\vec{e}_3]$$

Now interpret $\vec{e}_1$ as a deflection vector. Then $D^{-1}\vec{e}_1$ lists the forces that create the deflection. That is, the first column of D^{-1} lists the forces that must be applied at the three points to produce a unit deflection at point 1 and zero deflections at the other points.

Similarly, columns 2 and 3 of D^{-1} lists the forces required to produce unit deflections at points 2 and 3, respectively.

2.2 Inverse of a matrix (cont'd)

IV

Example 3 (cont'd)

In each column, one or two of the forces must be negative (point upward) to produce a unit deflection at the desired point, and zero deflections at the other two points. If the flexibility is measured, for example, in inches of deflection per pound of load, then the stiffness matrix entries are given in pounds of load per inch of deflection.

$$\frac{\text{in}}{\text{lb}} \quad \frac{\text{lb}}{\text{in}}$$

The formula in theorem 5 is seldom used to solve an equation $A\vec{x} = \vec{b}$ numerically because row reduction of $[A \mid \vec{b}]$ is nearly always faster.

Example 4

Use the inverse of the matrix A in example 2 to solve the system

$$3x_1 + 4x_2 = 3$$

$$5x_1 + 6x_2 = 7$$

Solution

This system is equivalent to $A\vec{x} = \vec{b}$, so

$\vec{x} = A^{-1}\vec{b}$, we know from example 2 that

$$A^{-1} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}$$

so

$$\vec{x} = A^{-1}\vec{b} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix} = 3 \begin{bmatrix} -3 \\ 5/2 \end{bmatrix} + 7 \begin{bmatrix} -3 \\ 5/2 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$$

2.2 The Inverse of a Matrix (cont'd)

V

Theorem 6

a. If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A$$

b. If A and B are $n \times n$ invertible matrices, then so is AB , and the inverse of AB is the product of the inverses of A and B in the reverse order. That is,

$$(AB)^{-1} = B^{-1}A^{-1}$$

c. If A is an invertible matrix, then so is A^T , and the inverse of A^T is the transpose of A^{-1} . That is,

$$(A^T)^{-1} = (A^{-1})^T$$

From example 2 we have $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}$

$$A^{-1}A = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Proof

To verify statement (a), find a matrix C such that

$$A^{-1}C = I \text{ and } CA^{-1} = I$$

Elementary Matrices

An elementary matrix is one that is obtained by performing a single elementary row operation on an identity matrix. The next example illustrates the three kind of elementary matrices.

Example 5

Let

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Compute E_1A , E_2A and E_3A , and describe how these products can be obtained by elementary row operations on A .

Solution

$$E_1A = \begin{bmatrix} a & b & c \\ d & e & f \\ g-4a & h-4b & i-4c \end{bmatrix}$$

$$E_2A = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix} \quad E_3A = \begin{bmatrix} a & b & c \\ d & e & f \\ 5g & 5h & 5i \end{bmatrix}$$

$$E_1A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Looking at rows, 100 means; 1 times row 1 ($a \ b \ c$) + 0 times row 2 ($d \ e \ f$) + 0 times row 3 ($g \ h \ i$). Now, row 3 in E_1 means -4 times row 1, 1 times row 3.

$E_{13}(E_1 \text{ row } 3)$ means $R_3 \leftarrow R_3 - 4R_1$

This means we can use elementary matrices to perform row operations. Multiplying several elementary matrices makes it possible to perform several row operation at once.

$$B = E_1A, \quad C = E_2B$$

$$E = E_2E_1 \Rightarrow D = EA \Rightarrow \text{equals } C$$

2.2 The Inverse of a matrix (cont'd)

VII

Example 6

Find the inverse of $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$

Solution

To transform E_1 into I , add 4 times row 1 to row 3. The elementary matrix that does this is

$$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$$

The following theorem provides the best way to "visualize" an invertible matrix, and the theorem leads immediately to a method for finding the inverse of a matrix.

Theorem 7

~~An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n , and in this case any sequence of elementary row operations that reduces A to I_n also transforms I_n into~~

Theorem 7

An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n , and in this case any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

P.140

An algorithm for finding A^{-1}

If we place A and I side by side in an augmented matrix $[A \ I]$, then performing row operations on this matrix will produce identical operations on A and I , eventually turning A into I and I into A^{-1} , as stated by theorem 7.

Algorithm for finding A^{-1}

Row reduce the augmented matrix $[A \ I]$. If A is row equivalent to I , then $[A \ I]$ is row equivalent to $[I \ A^{-1}]$. Otherwise A does not have an inverse.

P.140

2.2 The Inverse of a matrix (cont'd)

VIII

Example 7

Find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \text{ if it exists.}$$

Solution

$$[A \ I] = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \leftarrow R_3 \cdot \frac{1}{2}$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_3$$

$$R_2 \leftarrow R_2 - 2R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix}$$

$$R_3 \leftarrow R_3 + 3R_2$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{bmatrix}$$

Theorem 7 shows that, since $A \sim I$, that

$$A \text{ is invertible and } A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}$$

Let's put it to the test.

$$A(A^{-1}) = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$